Formation of Ionic Compounds by Burning Magnesium Ribbon

Introduction: We burned magnesium to create an ionic compound. Ionic compounds are when 2 ionic elements(1 cation and 1 anion) form a compound. A way to check that it is an ionic compound is if it conducts electricity. Magnesium reacts with certain components of air, this being nitrogen and oxygen that then form magnesium oxide and magnesium nitride.

Experimental Method:

Materials:
magnesium ribbon,15cm
evaporating dish
ring stand and ring
clay triangle
Bunsen burner
crucible tongs
balance
250 mL beaker
distilled water,11ml
dropper
scoopula
conductivity tester

Procedure:
1. Arrange ring on ring stand 7 cm above Bunsen burner. Place clay triangle on ring.
2. Roll 15cm of magnesium into loose ball and place in evaporating dish.
3. Place evaporating dish on clay triangle.
4. Turn flame on, place magnesium ribbon into flame with crucible tongs until it burns then place into evaporating dish, turn flame off.
5. Take a dropper after cooling and place one drop of distilled water in dish until reaction stops.
6. Turn flame on, heat evaporating dish until all the water evaporates.
7. Turn flame off, wait for crucible to cool.
8. Use scoopula to scrape magnesium product into beaker.
9. Add 10mL of distilled water into beaker and stir.
10. Use conductivity tester to test for ionic compounds.

Data:

Data: Masses of materials

Material(s)	**Mass (g)**

Empty evaporating dish	38.68
Mg ribbon and evaporating dish (together) before heating	38.87
Magnesium ribbon	.19
Magnesium products and evaporating (together) after heating	38.91
Magnesium product	.23

Conclusion:

We were able to successfully create an ionic compound during our lab and we can prove this because our conductivity tester came back positive on the “2” mark. The mistake that we possibly made was directly putting the burning magnesium when it was lit instead of waiting for it to be cooled but the product still came out to be ionic. The only way to prevent this was by waiting for the cooling to happen. The most fascinating part was when the burning of magnesium occurred. We also compared to other results and our results came out to be fairly average with some having a conductivity of “3” and we didn't have any expected value but we only expected for it to conduct some electricity and that is what happened. The reaction did cause us to gain mass from the beginning amount from .19 to .23 a .04 gram increase.

Discussion:

a) **Write the electron configuration for magnesium.**
   - **Based on the configuration, will magnesium lose or gain electrons to become a magnesium ion?**
   - **Write the electron configuration of a magnesium ion.**
   - **What noble gas has the same configuration as magnesium ion?**

$1s^2 2s^2 2p^6 3s^2$ It will lose electrons since it only needs to lose 2 to become a full shell Neon as it would be [Ne} $3s^2$

b) **Repeat a) above for oxygen.**
$1s^2 2s^2 2p^4$

c) **Repeat a) above for nitrogen.**
$1s^2 2s^2 2p^4$

d) **How do you know that the magnesium reacts with certain components of air?**
Magnesium Nitride and Magnesium Oxide Nitrogen and Oxygen are in the air all around us and since it can react with them that's how we know it can react with some of the components in the air.

e) **Magnesium reacts with oxygen and nitrogen in the air at high temperatures. Predict the molecular formulas for the products. Write the names of these compounds.**
Magnesium Nitride - Mg_3N_2 and Magnesium Oxide - MgO

f) The product of the magnesium and oxygen is white and the product of magnesium and nitrogen is yellow. Which one dominates your product?

White was the color so it most likely was Magnesium Oxide.

g) Did the product produce an electric current? Does this indicate whether or not the compounds are ionic?

Yes my product did produce an electric current because we tested it with a tester and the light for electric current came on meaning that the compounds are ionic.